sonam — -zsh — 80×24

```
sonam@Sonams-MacBook-Pro ~ % pip install numpy
zsh: command not found: pip
sonam@Sonams-MacBook-Pro ~ % pip3 install numpy
Requirement already satisfied: numpy in /Library/Frameworks/Python.framework/Ver
sions/3.11/lib/python3.11/site-packages (2.4.3)

[notice] A new release of pip available: 22.3.1 -> 26.0.1
[notice] To update, run: pip3 install --upgrade pip
sonam@Sonams-MacBook-Pro ~ % y
zsh: command not found: y
sonam@Sonams-MacBook-Pro ~ % pip3 install --upgrade pip
Requirement already satisfied: pip in /Library/Frameworks/Python.framework/Versi
ons/3.11/lib/python3.11/site-packages (22.3.1)
Collecting pip
  Downloading pip-26.0.1-py3-none-any.whl (1.8 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 10.3 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 22.3.1
    Uninstalling pip-22.3.1:
      Successfully uninstalled pip-22.3.1
Successfully installed pip-26.0.1
sonam@Sonams-MacBook-Pro ~ %
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
- External Libraries
- Scratches and Consoles

Task 1_1.py

```python
#your answer
#task1_1.py
import numpy as np
arr = np.arange( start_or_stop: 10,  stop: 20)
print("Array:", arr)
print("Shape:", arr.shape)
print("Dimension:", arr.ndim)
print("Data type:", arr.dtype)
```

Run Task 1_1

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_1.py
Array: [10 11 12 13 14 15 16 17 18 19]
Shape: (10,)
Dimension: 1
Data type: int64

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py

```python
#your answer
#task1_2.py
import numpy as np
arr = np.zeros( shape: (3, 3), dtype=int)
arr[1, 1] = 7
print(arr)
```

Run Task 1_2

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_2.py
[[0 0 0]
 [0 7 0]
 [0 0 0]]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py

```python
#your answer
#task1_3.py
import numpy as np
arr = np.array([[10, 20, 30],
[40, 50, 60],
[70, 80, 90]])
print("Second row:", arr[1])
print("Third column:", arr[:, 2])
print("Submatrix (first 2 rows and cols):")
print(arr[:2, :2])
```

Run Task 1_3

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_3.py
Second row: [40 50 60]
Third column: [30 60 90]
Submatrix (first 2 rows and cols):
[[10 20]
 [40 50]]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py Task 1_4.py

```python
#your answer
#task1_4.py
import numpy as np
arr = np.ones( shape: (2, 2, 3), dtype=int)
arr[1, 0, 2] = 5
print(arr)
```

Run Task 1_4

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_4.py
[[[1 1 1]
  [1 1 1]]

 [[1 1 5]
  [1 1 1]]]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py Task 1_4.py Task 1_5.py

```python
#your answer
#task1_5.py
import numpy as np
arr = np.array([2, 4, 6, 8, 10])
print(arr[::2])
```

Run Task 1_5

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_5.py
[ 2  6 10]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py Task 1_4.py Task 1_5.py Task 1_6.py

```python
#your answer
import numpy as np
a = np.array([5, 10, 15])
b = np.array([2, 4, 6])
print("Sum:", a + b)
print("Difference:", a - b)
print("Product:", a * b)
print("Division:", a / b)
```

Run Task 1_6

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_6.py
Sum: [ 7 14 21]
Difference: [3 6 9]
Product: [10 40 90]
Division: [2.5 2.5 2.5]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
- External Libraries
- Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py

```python
#your answer
import numpy as np
matrix = np.arange( start_or_stop: 1,  stop: 10).reshape(3, 3)
vector = np.array([1, 0, -1])
result = matrix * vector
print(result)
```

Run Task 1_7

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_7.py
[[ 1  0 -3]
 [ 4  0 -6]
 [ 7  0 -9]]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_b
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py

```python
#your answer
import numpy as np
arr = np.arange( start_or_stop: 0,  stop: 16,  step: 3)
sqrt_arr = np.sqrt(arr)
print(sqrt_arr)
```

Run Task 1_8

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_8.py
[0.         1.73205081 2.44948974 3.         3.46410162 3.87298335]

Process finished with exit code 0
```

python_basics Version Control

Project

- PythonProject [python_b
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py | Task 1_2.py | Task 1_3.py | Task 1_4.py | Task 1_5.py | Task 1_6.py | Task 1_7.py | Task 1_8.py | Task 1_9.py

```python
#your answer
import numpy as np
temps = np.array([
 [[10, 15, 20], [25, 30, 35]],
 [[11, 16, 21], [26, 31, 36]]
])
```

Run Task 1_9

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_9.py

Process finished with exit code 0
```

python_basics Version Control Current File

Project

- PythonProject [python_bas
  - .venv
  - Week4LabActivity
  - Week5LabActivity
    - Task 1_1.py
    - Task 1_2.py
    - Task 1_3.py
    - Task 1_4.py
    - Task 1_5.py
    - Task 1_6.py
    - Task 1_7.py
    - Task 1_8.py
    - Task 1_9.py
    - Task 1_10.py
  - External Libraries
  - Scratches and Consoles

Task 1_1.py Task 1_2.py Task 1_3.py Task 1_4.py Task 1_5.py Task 1_6.py Task 1_7.py Task 1_8.py Task 1_9.py Task 1_10.py

```python
#your answer
import numpy as np

# Given data
growth_factors = np.array([0.7, 0.2, 0.5])

rainfall = np.array([
    [40, 50, 60],
    [20, 35, 25],
    [30, 40, 55]
])

# Calculate height increase using broadcasting
height_increase = growth_factors[:, np.newaxis] * rainfall

# Print result
print(height_increase)
```

Run Task 1_10

```
/Users/sonam/PycharmProjects/PythonProject/.venv/bin/python /Users/sonam/PycharmProjects/PythonProject/Week5LabActivity/Task 1_10.py
[[28.  35.  42. ]
 [ 4.   7.   5. ]
 [15.  20.  27.5]]

Process finished with exit code 0
```